

NTMS Terraform D-1

14 Questions

1. What are some advantages of using Infrastructure as Code in an organization? (select three)

- 0/0 ☒ A IaC code can be used to manage infrastructure on multiple cloud platforms.
- 0/0 ☒ B IaC allows you to save your configurations in version control, enabling safe collaboration on infrastructure.
- 0/0 ☒ C IaC uses a human-readable configuration language to help you write infrastructure code quickly.
- 0/0 ☐ D IaC is written as an imperative approach, where specific commands need to be executed in the correct order.

2. Which of the following statements is the most accurate about the Terraform language?

- 0/0 ☐ A Terraform is an immutable, imperative, Infrastructure as Code configuration management language based on HashiCorp Configuration Language, or optionally JSON.
- 0/0 ☐ B Terraform is a mutable, imperative, Infrastructure as Code provisioning language based on HashiCorp Configuration Language, or optionally YAML.
- 0/0 ☒ C Terraform is an immutable, declarative Infrastructure as Code language based on HashiCorp Configuration Language or JSON.
- 0/0 ☐ D Terraform is a mutable, declarative, Infrastructure as Code configuration management language based on HashiCorp Configuration Language, or optionally JSON.

3. A development team is deciding between two deployment strategies: either using Terraform to provision new infrastructure for each deployment or provisioning base infrastructure once and updating it with configuration changes. Which of the following are advantages of the immutable infrastructure pattern? (select three)

- 0/0 ☐ A requires less storage space since infrastructure definitions are smaller than application artifacts
- 0/0 ☒ B eliminates configuration drift since servers are never modified after initial provisioning
- 0/0 ☐ C eliminates the need for configuration management tools in the deployment pipeline
- 0/0 ☒ D reduces the risk of inconsistent environments resulting from configuration updates over time
- 0/0 ☒ E simplifies rollback procedures by deploying previous infrastructure versions rather than reverting changes

4. Which of the following are advantages of using Infrastructure as Code (IaC) compared to traditional infrastructure management approaches? (select three)

- 0/0 ☒ A the same configuration can be consistently applied across multiple environments
- 0/0 ☐ B Infrastructure as Code eliminates human errors in infrastructure management
- 0/0 ☒ C infrastructure configurations can be versioned and stored in source control repositories
- 0/0 ☐ D it requires less initial setup time than using cloud provider UI consoles
- 0/0 ☐ E IaC tools automatically optimize infrastructure costs without human intervention
- 0/0 ☒ F teams can review and collaborate on infrastructure changes before they are applied

5. A startup needs to deploy its application across AWS, Google Cloud, and an on-premises VMware environment. Which advantages does Terraform provide for this multi-cloud setup? (select three)

- 0/0 ☐ A automatically replicate all resources from one cloud to another cloud
- 0/0 ☒ B write infrastructure configurations in one consistent language (HCL) for all providers
- 0/0 ☒ C use the same commands (init, plan, apply) regardless of the target platform
- 0/0 ☐ D convert existing CloudFormation templates directly into ARM templates
- 0/0 ☐ E eliminate the need for provider-specific authentication and credentials
- 0/0 ☒ F manage dependencies between resources across different cloud providers

6. Why is using a single tool like Terraform for multi-cloud deployments more beneficial than using separate tools and workflows for each cloud?

- 0/0 ☐ A It eliminates the need for cloud-specific credentials, since a single token works across all providers by default.
- 0/0 ☐ B It automatically migrates existing resources provisioned from other tools into Terraform without refactoring.
- 0/0 ☒ C It provides a common workflow and reusable modules, enabling consistent CI/CD and policy across clouds.
- 0/0 ☐ D It guarantees identical resource types and APIs across all clouds, eliminating any provider-specific differences.

7. What sets Infrastructure as Code (IaC) apart from managing infrastructure directly instead of making raw API calls or executing CLI commands?

- ☐ **A** Terraform keeps infrastructure logic in version control to ensure operations remain imperative and run step-by-step.
- ☒ **B** Terraform uses declarative configuration to describe the desired end state and generates a plan of action before applying changes.
- ☐ **C** Terraform guarantees that resources consistently reflect the desired state by automatically updating them as needed.
- ☐ **D** Terraform replaces provider APIs with its own protocol, removing the need to authenticate to each cloud.

8. What are the advantages of using Infrastructure as Code?

- ☒ **A** Infrastructure as Code is easily repeatable, allowing the user to reuse code to deploy similar yet different resources.
- ☒ **B** Infrastructure as Code enables users to automate a manual task for easier deployment.
- ☒ **C** Infrastructure as Code gives the user the ability to recreate an infrastructure for disaster recovery scenarios
- ☒ **D** Infrastructure as Code provides configuration consistency and standardization among deployments.
- ☒ **E** Infrastructure as Code is relatively easy to learn and write, regardless of a user's prior coding experience.

9. What benefits would you see by using a multi-cloud and provider-agnostic tool like Terraform? (select two)

- ☐ **A** Removes the need for provider-specific credentials or authentication flows.
- ☒ **B** Delivers a consistent declarative workflow and language across providers and hypervisors.
- ☒ **C** Reduces operational overhead by allowing teams to learn and govern a single tool across all environments.
- ☐ **D** Programmatically standardizes pricing and usage models across all cloud platforms.

10. Management wants to understand how adopting Infrastructure as Code with Terraform differs from your current method of using the console and CLI to deploy and manage infrastructure. Which statement correctly identifies a major difference?

- ☐ **A** Infrastructure as Code eliminates the need for cloud provider APIs since all infrastructure is declared in code.
- ☐ **B** Infrastructure as Code only works with immutable infrastructure patterns, while manual approaches support both mutable and immutable infrastructure.
- ☒ **C** Infrastructure as Code allows infrastructure to be described using a configuration syntax that can be versioned, reused, and shared.
- ☐ **D** Infrastructure as Code requires all resources to be created through APIs instead of UI consoles, while Terraform can work with both approaches.

- 11.** Your team manages long-lived VMs with a configuration management tool, but drift and rollbacks are frequent. You're evaluating alternatives. Which option best reflects an IaC advantage over traditional configuration management?
- 0/0** **A** IaC isn't useful in this scenario. Standardize on imperative scripts to run identical steps on each server during scheduled maintenance windows.
 - 0/0** **B** Use configuration management to provision resources and IaC to patch resources while keeping the same mutable servers and workflows.
 - 0/0** **C** Maintain configuration management for servers and export the current VMs as golden images once configurations have converged.
 - 0/0** **D** Using declarative IaC and immutable infrastructure allows you to define the desired state in code, easily replace resources, and version everything.
- 12.** Which characteristics are commonly associated with Infrastructure as Code (IaC) workflows?
- 0/0** **A** managing and provisioning infrastructure using machine-readable definitions stored in version control.
 - 0/0** **B** a tool that automatically resolves production incidents without requiring human intervention.
 - 0/0** **C** a GUI workflow for creating cloud resources via wizards.
 - 0/0** **D** a methodology where infrastructure changes are requested via service tickets and applied manually.
- 13.** Your organization uses Chef to configure application settings and install software packages on existing servers. The DevOps team is evaluating whether to adopt Terraform alongside Chef. Which statement best explains how Infrastructure as Code tools, like Terraform, differ from configuration management tools?
- 0/0** **A** Infrastructure as Code tools focus on provisioning and managing the lifecycle of infrastructure resources, while configuration management tools focus on installing and managing software on existing servers.
 - 0/0** **B** Infrastructure as Code tools can only create cloud resources and virtual machines, while configuration management tools can create both cloud and on-premises resources, including physical hardware.
 - 0/0** **C** Infrastructure as Code tools use agentless architectures to communicate with cloud providers, while configuration management tools always require agents to be installed on target systems.
 - 0/0** **D** Infrastructure as Code tools use imperative programming models to define step-by-step infrastructure changes, while configuration management tools use declarative models for desired state.

14. Sanjay has written a single Terraform configuration and wants to use it to deploy resources for his organization. Which statements about this configuration can be true? (select three)

- 0/0 ☐ **A** Each provider requires a separate configuration file since a single configuration cannot declare more than one provider.
- 0/0 ☒ **B** Its state can be stored in one cloud's remote backend while it provisions resources in another.
- 0/0 ☐ **C** The selected backend determines resource creation, so storing state in Azure limits it to only provisioning Azure resources.
- 0/0 ☒ **D** The same configuration can deploy QA and Staging by supplying different variable values or files
- 0/0 ☒ **E** It can manage resources across multiple cloud providers in the same run when both providers are configured.